Reproducibility Report for 'Vocal repertoire expansion in singing mice by co- opting a conserved midbrain circuit node' by Zheng et al. (2025, Current Biology)

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Introduction

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Justification for choice of study

I am choosing this study because it applies to my own research and is also incredibly well documented, so I think it will fit nicely for this project. It involves a lot of different interesting findings/experiments that concern the neural mechanisms of social behavior in a species of mouse. Specifically, I want to replicate the finding that cIPAG inhibition causes deficits in vocal behavior such that male mice included in the study made progressively shorter calls over the course of several days. These findings are summarized in figure 7 of the paper. Specifically, I would like to replicate the finding that song duration is significantly decreased in post-late injection condition compared to pre-injection. I think the data looks really nice and I appreciate the intervention method.

Anticipated challenges

I think it will be difficult for me to convert code from python into R. All of the author's analysis is done via jupyter notebooks, but I would like to do the project in R, so I need to be able to understand how to understand their code and translate it similarly into R. I also think it will be difficult to simulate data at the correct level, instead of simulating raw data I will be simulating the time-stamped vocal behavior (as I can't simulate video/audio files of behavior). This is a pretty large amount of data that is all separated into different files, looking at the data they provided. I will need to make sure the organization of this data is clear and manageable the way that they did.

Currently, I believe the most challenging part is extracting "song duration" in the same way the authors did. This involves not only filtering out all notes of type "song" made by male mice, but linking these song notes together into song durations. I am working on understanding how they quantify songs in their code so I can use the same parameters in my analysis (see current progress in jupyter_translate on this repo). After translating everything into R, I can conduct enough analyses of the data to understand how to simulate it for my own study.

Links

Project repository (on Github): repo

Original paper (as hosted in your repo): paper

Methods

In this paper, they tested whether inhibition of the brain area cIPAG causes a deficit in singing behavior in male singing mice. They recorded audio from 5 male mice during behavioral sessions in which they could interact with a female conspecific (social context which normally prompts singing vocalizations). They recorded vocalizations pre and post-injection of

a chemogenetic inhibitor into cIPAG and analyzed this audio using a model which categorized vocalizations into individual notes with timestamps and vocalization type.

Description of the steps required to reproduce the results

I have downloaded the data files for this experiment from the resource provided in the paper. I can run analyses on song duration for each mouse with respect to its condition (time from inhibitory injection) across sessions to get a statistical summary of vocal behavior for each participant. Then, I will simulate a data consisting of individual call durations during each session for each of 5 mice. The number of behavioral observations varies in each session, so I will produce data observations within a range. Then, I will compare average song durations between time points, and check if there is a significant difference in call duration between each timepoint. The authors used Mann-Whitney U-tests for this.

Differences from original study

I will be using R instead of python to conduct my analysis. Also I will be simulating semi-processed data instead of the raw behavioral data.

Project Progress Check 1

Measure of success

I will be replicating the Mann-Whitney test comparing pre-injection to post-late ($p = 1.53e - 19$). I would consider the simulation successful if I find a similarly significant p-value (after using the same method/parameters for determining song duration using simulated notes per session). I will be comparing song duration in seconds across day 1 and day 5.

Pipeline progress

I am currently still in the process of understanding how they did the data analysis themselves. It is hard for me to understand exactly what they did because I don't read python very well. So far, I have loaded their data into R (all vocal notes made by all mice in all sessions), and filtered out only male song notes and done some very basic plotting. I have found that song note duration follows the same trend in decreasing across conditions for all mice. I have been attempting to compare average song duration (replicating same figure with the same data) but I am struggling to translate the exact methods into R from jupyter notebook. The findings are still the same according to the plots, but the average song durations are much shorter in my code compared to the way they calculate song duration. I am working on getting the parameters right so that I am capturing song duration accurately from individual notes.

Results

Data preparation

Data preparation following the analysis plan.

Key analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Reproduction Attempt

Open the discussion section with a paragraph summarizing the primary result from the key analysis and assess whether you successfully reproduced it, partially reproduced it, or failed to reproduce it.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis of the dataset, (b) assessment of the meaning of the successful or unsuccessful reproducibility attempt - e.g., for a failure to reproduce the original findings, are the differences between original and present analyses ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the reproducibility attempt (if you contacted them). None of these need to be long.